SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
Student Name:		**Reg. No.:**	
Date:		**Duration:**	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario**
   - Carefully understand the given problem statement.
   - Identify the business domain, stakeholders, existing challenges, and system objectives.
2. **Identify Functional and Non-Functional Requirements**
   - List all functional requirements required by the proposed system.
   - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
3. **Identify Actors and Use Cases**
   - Identify all primary and secondary actors interacting with the system.
   - List the major use cases associated with each actor.
   - Draw a Use Case Diagram representing the interactions between actors and the system.
4. **Prepare the Software Requirement Specification (SRS)**
   Develop an SRS document containing:
   - Introduction
   - Problem Description
   - Scope of the System

- Functional Requirements
- Non-Functional Requirements
- Use Case Descriptions
- Data Requirements
- External Interface Requirements
- System Features

5. **Identify Assumptions and Constraints**
   - List the assumptions considered while developing the system.
   - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
6. **Validate Requirement Completeness and Consistency**
   - Verify whether all stakeholder requirements have been addressed.
   - Check for ambiguity, redundancy, inconsistency, and missing requirements.
   - Suggest improvements to enhance the quality and completeness of the requirement specification.

1. Study and Analysis of the Scenario

1.1 Problem Statement

A multi-specialty hospital with several hundred outpatients per day still manages appointments through telephone calls, walk-in queues and paper registers. Patients wait for long hours, doctors' time slots are double-booked, patient history is scattered across paper files, billing is done manually, and the management has no reliable data about doctor workload or revenue. The hospital needs a web and mobile based system that allows patients to book, reschedule and pay for appointments online, supports doctors in recording consultations and prescriptions, helps the front desk with check-in and billing, and gives the administration accurate reports.

1.2 Business Domain

The system belongs to the Healthcare Information Systems (Health-Tech) domain, specifically Outpatient Department (OPD) management. It deals with sensitive personal and medical data, so privacy, accuracy and availability are the most important quality concerns.

1.3 Stakeholders

Stakeholder	Role / Interest	Key Expectation
Patient	Primary user who books and pays for care	Quick booking, no waiting, access to own reports and prescriptions
Doctor	Provides consultation	Clear daily schedule, quick access to history, easy prescription entry
Receptionist	Front-desk operations	Fast walk-in booking, check-in and billing
Administrator	Controls the system	User and doctor management, reports, audit control
Hospital Management	Business owner	Higher patient throughput, revenue visibility, reduced paperwork
Payment Gateway / Bank	External financial service	Secure and reliable transaction exchange
SMS / Email Provider	External messaging service	Correct API usage and message delivery
Government / Regulator	Compliance authority	Data protection and medical record retention rules

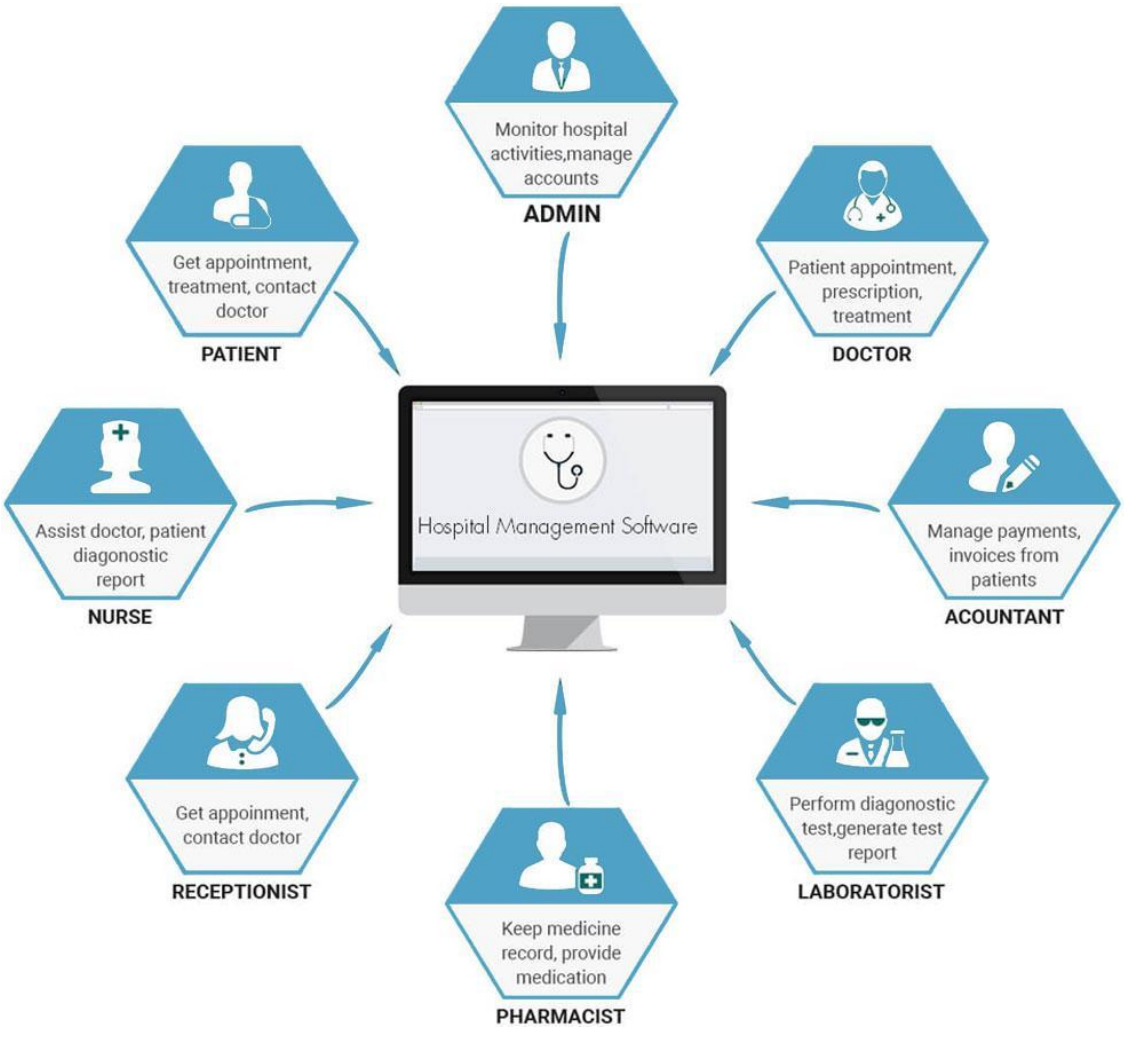
1.4 Existing Challenges

- **Long waiting time:** patients queue physically to get a token or an appointment.

- **Double booking and no-shows:** manual registers cannot prevent two patients in one slot or remind absentees.
- **Fragmented records:** paper files get lost, are illegible, and cannot be shared between departments.
- **Manual billing errors:** calculation mistakes and delayed payment reconciliation.
- **Poor visibility:** management cannot see doctor utilisation, patient load or revenue trends.
- **Weak security and privacy:** paper records are open to unauthorised viewing and there is no audit trail.

1.5 System Objectives

- Provide 24 × 7 online appointment booking, cancellation and rescheduling.
- Reduce average patient waiting time and eliminate double booking.
- Maintain a centralised, secure electronic medical record for every patient.
- Automate billing, online payment and receipts.
- Send automatic reminders and status notifications to patients and doctors.
- Give management real-time dashboards and exportable reports.
- Protect patient data through role-based access, encryption and audit logs.



2.Introduction (SRS)

2.1 Purpose

This document specifies the software requirements for the Hospital Appointment and Patient Management System (HAPMS). It is intended for the development team, testers, hospital management and the course evaluator, and acts as the baseline agreement between stakeholders and developers.

2.2 Definitions, Acronyms and Abbreviations

Term	Meaning
SRS	Software Requirement Specification
OPD	Outpatient Department
EMR	Electronic Medical Record
OTP	One-Time Password
RBAC	Role-Based Access Control
API	Application Programming Interface
UC / FR / NFR	Use Case / Functional Requirement / Non-Functional Requirement
UPI	Unified Payments Interface

2.3 References

- IEEE Std 830-1998 – Recommended Practice for Software Requirements Specifications.
- ISO/IEC/IEEE 29148:2018 – Requirements Engineering.
- I. Sommerville, Software Engineering, Pearson.
- R. S. Pressman, Software Engineering: A Practitioner’s Approach, McGraw-Hill.
- Digital Personal Data Protection Act, 2023 (India).

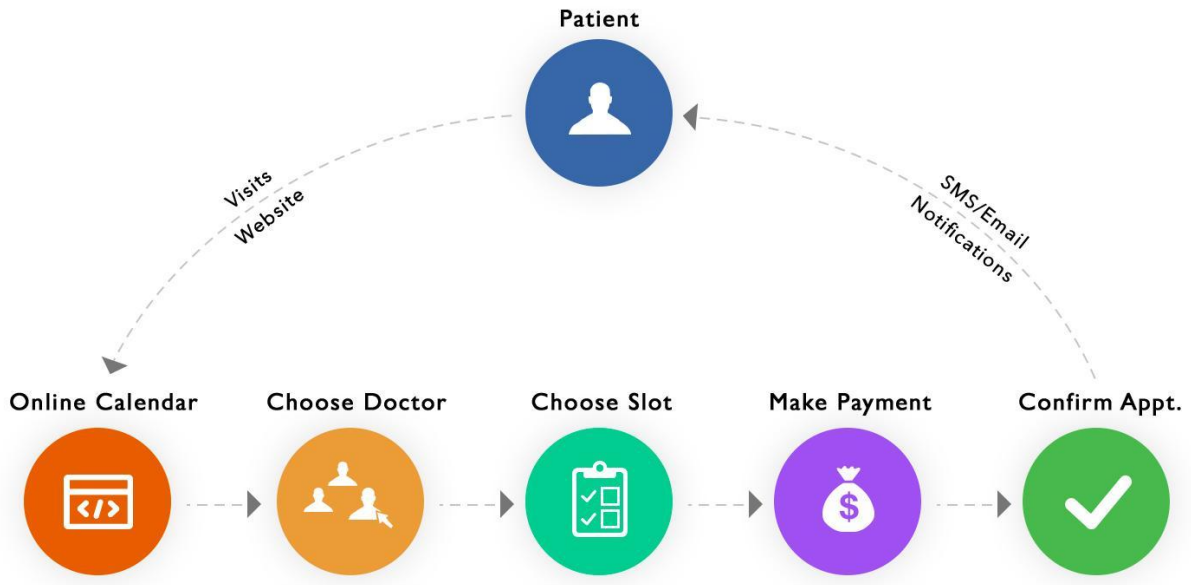
2.4 Document Overview

Sections 3 to 5 describe the problem, scope and requirements. Sections 6 and 7 present actors, the use case diagram and use case descriptions. Sections 8 to 10 cover data, interfaces and system features. Sections 11 and 12 list assumptions, constraints and the validation of the specification.

3. Problem Description

The present manual process causes appointment conflicts, long queues, loss of patient history and delayed billing. Because there is no central database, doctors cannot view previous visits and management cannot measure performance. HAPMS solves this by replacing the manual process with a single integrated, role-based platform that covers the complete outpatient

workflow: registration, doctor search, booking, check-in, consultation, prescription, billing, payment, notification and reporting.



4. Scope of the System

4.1 In Scope

- Patient registration, login and profile management.
- Doctor search by department, name and availability.
- Appointment booking, cancellation, rescheduling and walk-in booking.
- Doctor schedule and leave management.
- Patient check-in, consultation notes and e-prescription.
- Billing, online payment, receipts and refunds.
- Electronic medical record and report viewing.
- SMS / email notifications and reminders.
- Administrative user management, dashboards and reports.

4.2 Out of Scope

- Inpatient (IPD) bed management and surgery scheduling.
- Pharmacy stock and laboratory instrument integration.
- Insurance claim processing and telemedicine video calls (future enhancement).
- Hospital payroll and human-resource management.

5. Functional Requirements

Priority is given using the MoSCoW scale: M = Must have, S = Should have, C = Could have.

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ID	Module	Requirement	Priority
FR-01	User Management	The system shall allow a patient to register using name, mobile number, email, date of birth and gender, with OTP verification.	M
FR-02	User Management	The system shall authenticate users with username/password and OTP, and apply role-based access (Patient, Doctor, Receptionist, Admin).	M
FR-03	User Management	The system shall allow users to update their profile and reset a forgotten password.	S
FR-04	Doctor Search	The system shall allow searching doctors by department, specialisation, name, language and availability.	M
FR-05	Doctor Search	The system shall display doctor profile, experience, consultation fee and available slots.	S
FR-06	Appointment	The system shall let a patient book an appointment in an available slot only; a booked slot shall not be offered again.	M
FR-07	Appointment	The system shall allow a patient to cancel or reschedule an appointment up to 2 hours before the slot.	M
FR-08	Appointment	The system shall allow the receptionist to book, modify or cancel appointments on behalf of walk-in patients.	M
FR-09	Appointment	The system shall generate a unique appointment ID and a token number for every confirmed booking.	M
FR-10	Schedule	The system shall allow a doctor to define working days, time slots, slot duration and leave.	M
FR-11	Schedule	The system shall automatically notify affected patients when a doctor’s schedule changes or a doctor is on leave.	S
FR-12	Check-in	The system shall allow the receptionist to mark a patient as arrived and maintain a live queue for each doctor.	M
FR-13	Consultation	The system shall allow the doctor to view the patient’s history and record symptoms, diagnosis and advice.	M
FR-14	Consultation	The system shall allow the doctor to issue an e-prescription and request lab tests.	M

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ID	Module	Requirement	Priority
FR-15	Consultation	The system shall allow the doctor to schedule a follow-up visit.	C
FR-16	Records	The system shall store all visits, prescriptions and uploaded reports in the patient's EMR.	M
FR-17	Records	The system shall allow a patient to view and download own prescriptions, reports and bills.	M
FR-18	Billing	The system shall generate a bill containing consultation fee, tests and taxes for each visit.	M
FR-19	Payment	The system shall support online payment through UPI, cards and net banking via the payment gateway.	M
FR-20	Payment	The system shall record cash payments entered by the receptionist and generate a receipt.	M
FR-21	Payment	The system shall process refunds for eligible cancellations and update payment status.	S
FR-22	Notification	The system shall send SMS/email for booking confirmation, reminder (24 hours and 2 hours before), cancellation and payment receipt.	M
FR-23	Feedback	The system shall allow a patient to rate a completed consultation and submit feedback.	C
FR-24	Administration	The system shall allow the administrator to add, edit, deactivate doctors, departments and staff accounts.	M
FR-25	Reporting	The system shall generate reports on daily appointments, doctor workload, revenue and cancellations, exportable to PDF and Excel.	M
FR-26	Audit	The system shall log every access to and change of patient data with user, time and action.	M
FR-27	Search	The system shall allow authorised staff to search a patient by ID, name or mobile number.	S
FR-28	Waitlist	The system shall allow a patient to join a waiting list for a full slot and notify when it becomes free.	C

Non-Functional Quality Requirements



1

SECURITY

Focuses on securing the data and actions in the system against possible malware attacks or unauthorized access.

2

PERFORMANCE

Defines how quickly or smoothly the system reacts to user interaction under a specific workload.

3

SCALABILITY

Stands for the system's elasticity to expand its processing capacities.

4

THROUGHPUT

Describes the number of messages or transactions the system can process simultaneously.

5

RELIABILITY

Defines the probability that the system will not fail for a specific time interval or the number of uses.

6

AVAILABILITY

Describes the time frame the system is accessible for a user.

7

COMPATIBILITY & PORTABILITY

These two represent the software's ability to interact with other systems.

8

MAINTAINABILITY

It is the time, ease, and flexibility with which software can be modified.

9

USABILITY & ACCESSIBILITY

They describe if the system is easy to learn and operate for all users, including groups with a wide range of capabilities.

6. Non-Functional Requirements

ID	Category	Requirement	Measure
NFR-01	Performance	Pages and search results shall load quickly under normal load.	95% of requests within 3 seconds; booking confirmed within 5 seconds
NFR-02	Scalability	The system shall support growth in users, doctors and branches without redesign.	Handle 5,000 concurrent users; horizontal scaling supported
NFR-03	Availability	The system shall be available throughout the day, with planned maintenance in off-peak hours.	99.5% monthly uptime
NFR-04	Reliability	No confirmed booking or payment shall be lost or duplicated, even on failure.	Daily backup; recovery point ≤ 15 min; recovery time ≤ 2 hours
NFR-05	Security	Data shall be protected in storage and transit, and access restricted by role.	HTTPS/TLS 1.2+, AES-256 at rest, hashed passwords, session timeout 15 min, account lock after 5 failed logins
NFR-06	Privacy / Legal	Medical data shall be used only for care and shall follow data protection law.	Consent capture; EMR retention as per law; audit logs retained 1 year
NFR-07	Usability	The interface shall be simple for first-time and elderly users and support English and Tamil.	A new patient books an appointment in ≤ 5 steps without training
NFR-08	Compatibility	The system shall work on common browsers and mobile devices.	Chrome, Edge, Firefox, Safari; Android 9+/iOS 14+; responsive layout
NFR-09	Maintainability	The code shall be modular and documented so that changes are easy.	Layered architecture; ≥ 70% unit-test coverage; defect fix cycle ≤ 2 days
NFR-10	Portability	The system shall be deployable on cloud or on-premise servers.	Containerised deployment; no vendor lock-in for the database
NFR-11	Accessibility	The interface shall follow accessibility guidelines.	WCAG 2.1 Level AA

ID	Category	Requirement	Measure
NFR-12	Data Integrity	Data shall remain consistent and accurate.	ACID transactions; slot uniqueness enforced at database level

7. Actors and Use Cases

7.1 Actors

Actor	Type	Description
Patient	Primary	Registers, searches doctors, books/cancels appointments, pays, views records and gives feedback.
Doctor	Primary	Manages schedule, views patient history, records consultation and issues prescriptions.
Receptionist	Primary	Handles walk-in booking, patient check-in and counter payments.
Administrator	Primary	Manages users, doctors, billing settings and reports.
Payment Gateway	Secondary	External system that processes online payments and refunds.
SMS / Email Service	Secondary	External service that delivers notifications and reminders.

7.2 Major Use Cases per Actor

Actor	Use Cases
Patient	UC-01 Login; UC-02 Search Doctors; UC-03 Book Appointment; UC-04 Cancel / Reschedule; UC-06 View Medical Records; UC-07 Give Feedback; UC-08 Make Online Payment
Doctor	UC-01 Login; UC-09 Manage Doctor Schedule; UC-10 Record Consultation; UC-11 Issue e-Prescription
Receptionist	UC-01 Login; UC-03 Book Appointment (walk-in); UC-04 Cancel / Reschedule; UC-05 Check-in Patient; UC-08 Make Payment (counter)
Administrator	UC-01 Login; UC-12 Manage Users & Doctors; UC-13 Generate Reports; UC-14 Manage Billing
Payment Gateway	UC-08 Make Online Payment (process transaction)

Actor	Use Cases
SMS / Email Service	UC-15 Send Notifications

7.3 Use Case Diagram

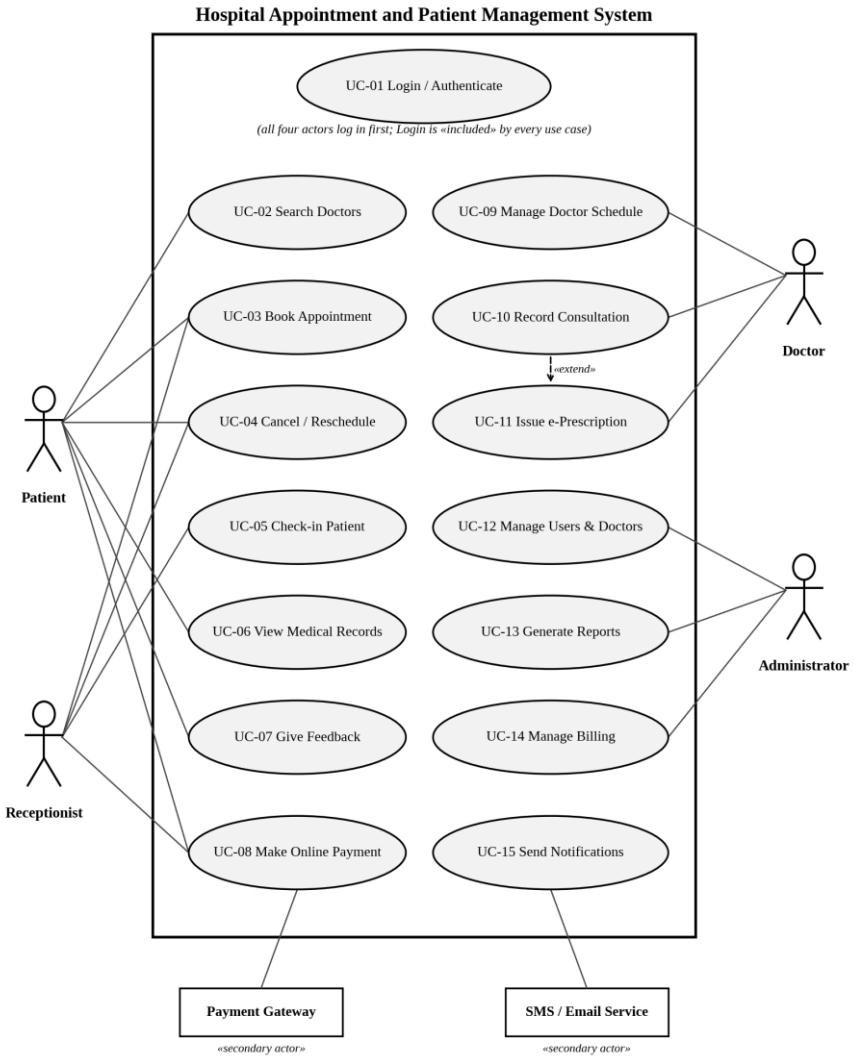


Figure 1: Use Case Diagram of the Hospital Appointment and Patient Management System

8. Use Case Descriptions

UC-01 – Login / Authenticate

Primary Actor	Patient, Doctor, Receptionist, Administrator
Goal	Give a registered user secure access to role-specific functions.
Precondition	User account exists and is active.

Main Flow	1. User opens the login page. 2. User enters username and password. 3. System validates credentials and sends an OTP. 4. User enters the OTP. 5. System verifies the OTP, starts a session and opens the role dashboard.
Alternate Flow	3a. Wrong credentials: system shows an error and counts the attempt. 4a. Wrong or expired OTP: user may request a new OTP.
Exception	After 5 failed attempts the account is locked for 30 minutes.
Postcondition	User is logged in; login event is written to the audit log.

UC-03 – Book Appointment

Primary Actor	Patient (Receptionist for walk-in patients)
Secondary Actor	SMS / Email Service
Goal	Reserve a consultation slot with a chosen doctor.
Precondition	User is logged in; doctor has at least one free slot.
Main Flow	1. Patient searches and selects a doctor (UC-02). 2. System displays the available dates and slots. 3. Patient selects a slot and enters the reason for the visit. 4. System locks the slot for 5 minutes. 5. Patient confirms the booking and chooses to pay now or at the counter. 6. System creates the appointment ID and token and marks the slot as booked. 7. System sends confirmation through SMS and email (UC-15).
Alternate Flow	3a. Slot taken by another user: system asks the patient to choose another slot or join the waiting list. 5a. Patient selects online payment: system runs UC-08.
Exception	Lock expires before confirmation: slot is released and the patient restarts selection.
Postcondition	Appointment is stored with status ‘Confirmed’; slot is unavailable to others.

UC-08 – Make Online Payment

Primary Actor	Patient
Secondary Actor	Payment Gateway

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Goal	Pay the consultation fee securely online.
Precondition	A bill or pending appointment exists.
Main Flow	1. System shows the bill amount and payment options. 2. Patient selects UPI, card or net banking. 3. System sends the request to the payment gateway. 4. Gateway authenticates and processes the payment. 5. System receives the success response, updates payment status and generates a receipt. 6. System sends the receipt to the patient.
Alternate Flow	4a. Payment fails: system shows the reason and allows retry or pay-at-counter.
Exception	Gateway timeout: the system checks transaction status before allowing a retry, to prevent double charging.
Postcondition	Payment is recorded as ‘Paid’ and linked to the appointment.

UC-10 – Record Consultation

Primary Actor	Doctor
Goal	Record the outcome of a consultation in the patient’s EMR.
Precondition	Patient is checked in; doctor is logged in.
Main Flow	1. Doctor selects the next patient from the queue. 2. System displays the patient’s profile, allergies and previous visits. 3. Doctor enters symptoms, diagnosis and advice. 4. Doctor optionally issues an e-prescription (UC-11) and orders tests. 5. Doctor saves and closes the consultation.
Alternate Flow	4a. Follow-up needed: doctor schedules a follow-up date.
Exception	Network failure: draft is auto-saved locally and synchronised when connection returns.
Postcondition	Visit record is saved in the EMR; appointment status becomes ‘Completed’.

UC-13 – Generate Reports

Primary Actor	Administrator
Goal	Obtain operational and financial reports.
Precondition	Administrator is logged in.

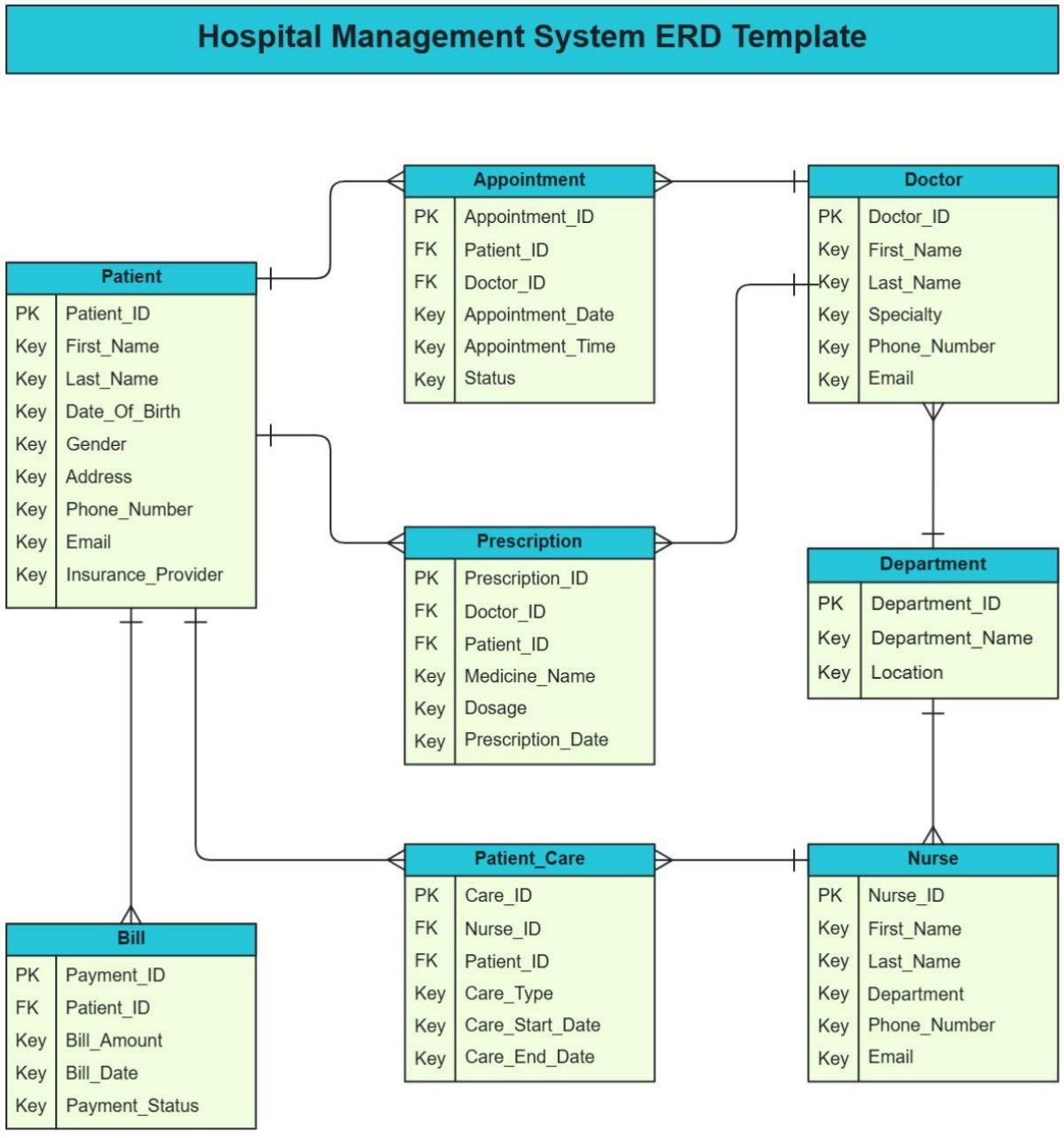
Main Flow	1. Administrator selects the report type. 2. Administrator chooses date range, department or doctor. 3. System collects data and displays the report with charts. 4. Administrator exports the report to PDF or Excel.
Alternate Flow	3a. No data for the filter: system shows ‘No records found’.
Postcondition	Report is generated; the export is logged.

9. Data Requirements

9.1 Major Data Entities

Entity	Key Attributes	Relationships
User	user_id (PK), username, password_hash, role, status, last_login	One user is one patient / doctor / staff profile
Patient	patient_id (PK), name, DOB, gender, mobile, email, address, blood_group, allergies	1 Patient – many Appointments, many Records
Doctor	doctor_id (PK), name, specialisation, qualification, experience, fee, dept_id (FK)	Belongs to 1 Department; has many Slots and Appointments
Department	dept_id (PK), name, description	1 Department – many Doctors
Schedule / Slot	slot_id (PK), doctor_id (FK), date, start_time, end_time, status	1 Doctor – many Slots; 1 Slot – 0..1 Appointment
Appointment	appt_id (PK), patient_id (FK), doctor_id (FK), slot_id (FK), token_no, status, reason	Links Patient, Doctor and Slot; has 1 Consultation and 1 Bill
Consultation	consult_id (PK), appt_id (FK), symptoms, diagnosis, advice, follow_up_date	1 Appointment – 1 Consultation
Prescription	presc_id (PK), consult_id (FK), medicine, dosage, duration, notes	1 Consultation – many Prescription lines
Bill / Payment	bill_id (PK), appt_id (FK), amount, tax, mode, txn_id, status, paid_on	1 Appointment – 1 Bill; 1 Bill – many Payments (refunds)
Medical Report	report_id (PK), patient_id (FK), type, file_path, uploaded_on	1 Patient – many Reports
Notification	notif_id (PK), user_id (FK), channel, message, sent_on, status	1 User – many Notifications

Entity	Key Attributes	Relationships
Feedback	feedback_id (PK), appt_id (FK), rating, comments	1 Appointment – 0..1 Feedback
Audit Log	log_id (PK), user_id (FK), action, entity, timestamp, ip_address	Records every sensitive action



9.2 Data Rules

- Mobile number and email shall be unique per patient; patient_id is system generated.
- A slot shall have at most one active appointment (unique constraint).
- Medical records shall never be physically deleted; only marked inactive.

- Payment data shall store the gateway transaction ID only; card numbers shall never be stored.
- Backups shall be encrypted and retained for 30 days; medical records are retained as per statutory period.

10. External Interface Requirements

Interface	Requirement
User Interface	Responsive web application and mobile app with role-based dashboards, forms with validation, clear error messages, and English / Tamil language option. Consistent layout, large fonts and high contrast.
Hardware Interface	Client: any device with a modern browser or smartphone. Server: application server and database server (minimum 8 GB RAM, 4 vCPU, SSD storage). Optional: barcode/QR scanner and receipt printer at the reception.
Software Interface	Relational database (MySQL / PostgreSQL); RESTful API between front end and back end; payment gateway API (UPI/cards/net banking); SMS and email gateway API; PDF/Excel export library.
Communication Interface	HTTPS (TLS 1.2+) for all traffic; JSON over REST; SMTP for email; secure webhook for payment callbacks.

11. System Features

Feature	Description	Related Requirements
F1 – Account & Access Management	Registration, OTP login, role-based access, profile and password handling.	FR-01 to FR-03, FR-26, NFR-05
F2 – Doctor Discovery	Search and filter doctors, view profile and live availability.	FR-04, FR-05
F3 – Appointment Management	Online and walk-in booking, cancellation, rescheduling, token generation, waiting list.	FR-06 to FR-09, FR-28
F4 – Schedule Management	Doctor slots, leave, automatic patient alerts.	FR-10, FR-11
F5 – Clinical Workflow	Check-in, live queue, consultation notes, e-prescription, follow-up.	FR-12 to FR-15
F6 – Electronic Medical Records	Central EMR, report upload and patient download.	FR-16, FR-17, FR-27

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Feature	Description	Related Requirements
F7 – Billing & Payment	Bill generation, online/cash payment, receipts, refunds.	FR-18 to FR-21
F8 – Notifications	SMS/email confirmation, reminders and alerts.	FR-22
F9 – Feedback	Rating and comments on consultation.	FR-23
F10 – Administration & Reporting	User/doctor management, dashboards, exportable reports, audit trail.	FR-24, FR-25, FR-26

12. Assumptions and Constraints

12.1 Assumptions

- Patients and staff have access to a smartphone or computer with internet connection.
- Every patient has a valid mobile number for OTP and notifications.
- Doctors provide their schedules in advance and update them when plans change.
- The hospital has a stable internet connection and a reliable power backup.
- The payment gateway and SMS/email provider offer stable APIs with agreed uptime.
- Staff will receive basic training before go-live.
- Existing paper records will be digitised gradually and are not part of the first release.
- Consultation fees are fixed per doctor and changed only by the administrator.

12.2 Constraints

Type	Constraint
Technical	Must integrate with third-party payment and messaging APIs; must run on standard browsers; limited bandwidth in rural areas requires light pages.
Operational	Reception staff must be able to work during peak hours with minimal training; system downtime is allowed only in off-peak maintenance windows.
Economic	Limited project budget; use open-source technologies (MySQL/PostgreSQL, web frameworks) and pay-per-use cloud and SMS services.
Legal	Compliance with the Digital Personal Data Protection Act, 2023, IT Act 2000 and medical record confidentiality; patient consent is mandatory; no card data storage (PCI-DSS handled by gateway).

Type	Constraint
Time	The first release must be completed within one academic semester (about 16 weeks): 3 weeks requirements, 4 weeks design, 6 weeks development, 3 weeks testing and deployment.

13. Validation of Requirement Completeness and Consistency

13.1 Stakeholder Coverage Check

Stakeholder Need	Covered By	Status
Patient – quick online booking	FR-04 to FR-09, UC-02, UC-03	Covered
Patient – access to own records and bills	FR-16, FR-17, UC-06	Covered
Doctor – schedule and history access	FR-10, FR-13, FR-14, UC-09, UC-10	Covered
Receptionist – walk-in handling and billing	FR-08, FR-12, FR-20, UC-05	Covered
Administrator – control and reports	FR-24, FR-25, FR-26, UC-12, UC-13	Covered
Management – revenue and workload visibility	FR-25, UC-13	Covered
Regulator – privacy and audit	FR-26, NFR-05, NFR-06	Covered
External systems – payment and messaging	FR-19, FR-22, UC-08, UC-15	Covered

13.2 Quality Check of the Requirements

Quality Attribute	Check Performed	Finding / Action Taken
Completeness	Every actor has at least one use case; every use case maps to a functional requirement.	No orphan use cases. Missing items found (refund, waiting list, audit) were added as FR-21, FR-28, FR-26.
Consistency	Terms, priorities and rules compared across sections.	Cancellation limit fixed at 2 hours in FR-07 and UC-04; slot uniqueness in FR-06 matches the data rule in Section 9.
Ambiguity	Vague words such as ‘fast’, ‘secure’, ‘user-friendly’ replaced.	Converted to measurable values in Section 6 (3 seconds, AES-256, 5 steps).

Quality Attribute	Check Performed	Finding / Action Taken
Redundancy	Duplicate statements searched.	Booking by patient and by receptionist merged into one use case with two actors; notifications defined once in FR-22.
Verifiability	Each requirement can be tested.	Each FR has an observable result; each NFR has a numeric measure.
Traceability	Requirement IDs mapped to use cases and features.	Mapping given in Sections 7, 8 and 11.
Feasibility	Technology and time checked.	Feasible with open-source tools and third-party APIs within one semester.

13.3 Suggested Improvements

- Prototype the booking and payment screens and review with real patients and reception staff to validate usability.
- Add a prioritised release plan: Release 1 – Must requirements; Release 2 – Should; Release 3 – Could.
- Define acceptance criteria and test cases for every Must requirement before development starts.
- Add integration with laboratory and pharmacy systems and insurance claims in a later release.
- Introduce a change-control process with version numbers so that new requirements do not create inconsistencies.
- Conduct a formal requirements review (walkthrough / inspection) with all stakeholders and sign off the baseline.

14. Conclusion

The analysis identified the stakeholders, challenges and objectives of the hospital scenario and specified 28 functional and 12 non-functional requirements, six actors and fifteen use cases with a UML use case diagram. The SRS covers all the required sections, lists realistic assumptions and constraints, and was validated for completeness, consistency, ambiguity and redundancy. Implementing this specification will reduce waiting time, avoid double booking, protect patient data and give the hospital reliable information for decision making.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Mark
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5

TOTAL	/25